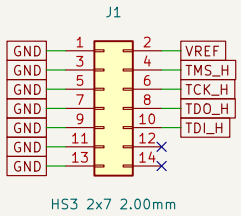
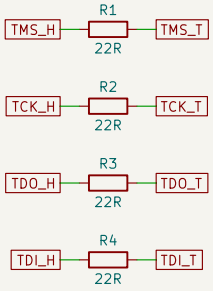
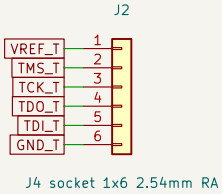
HS3 SIDE: Xilinx 14-pin JTAG header.
Pin 2 is the HS3 buffer supply (VREF).



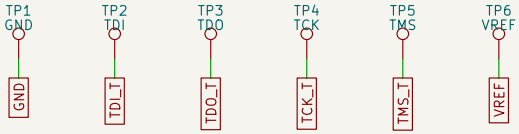
SERIES RESISTORS
(the HS3 already has 100R inside)



CARD SIDE: J2 plugs straight onto card header J4,
or takes a 6-way male-to-female jumper cable.
J2 pin n meets J4 pin n.



TEST HOLES (not fitted)
for a scope or logic analyzer

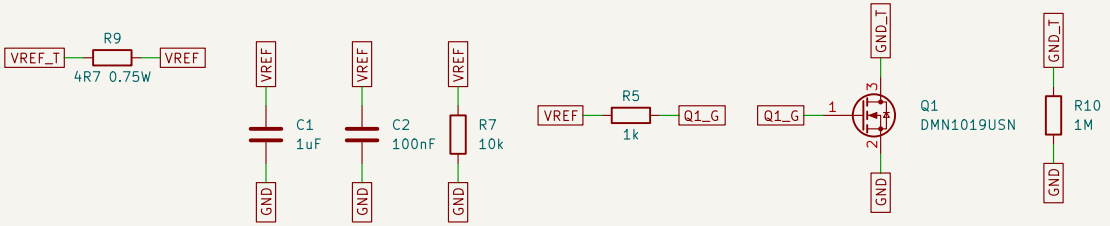


Card header J4 (AS02MC04, XCKU3P bank 0, 1.8 V ONLY)
pin 1 1V8 reference voltage from the card
pin 2 TMS
pin 3 TCK (silkscreen on the card says TDK)
pin 4 TDO
pin 5 TDI
pin 6 GND

1.8 V FEED: R9 limits what a fault behind it can draw
from the card's 1.8 V rail to 0.38 A. It is a 0.75 W part: a dead short
makes it hot, not burnt. Only J2 pin 1, the 2 mm link and R9 pad 1 are ahead of it.

REVERSE-INSERTION PROTECTION: Q1 sits in the ground lead.
Correct way round: gate sees 1.8 V, Q1 is on.
Reversed: gate sees 0 V, body diode blocks, no short.

REVERSED INDICATOR: sits across the two end pins (behind R9),
so it glows only when 1.8 V arrives the wrong way round.
Dim by nature (1.8 V). No light does NOT prove correct orientation.



Probe: Digilent JTAG-HS3 rev A, Xilinx 14-pin 2.00 mm header
Card: Alibaba Cloud AS02MC04 (PN YZQT-02337-101), header J4, 1.8 V JTAG

Sheet: /
File: as02mc04_jtag_hs3.kicad_sch

Title: AS02MC04 JTAG adapter for Digilent JTAG-HS3

Size: A4	Date: 2026-11-11	**Rev: C**
KiCad E.D.A. 10.0.8		Id: 1/1